

BAO-NGOC PHAN

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EDUCATION

Le Hong Phong High School for the Gifted

Specialization: Chemistry

GPA: 9.5/10.0 · AP Chemistry 5 · AP Calculus AB & BC 5

Ninh Binh, Vietnam

Aug 2023 – Jul 2026

- Achieved the only perfect 10 in specialized Chemistry in the 30-year history of the High School Entrance Exam.
- Received certificates of Merit from both the Nam Dinh City Youth Union and Nam Dinh Provincial Department of Education and Training for outstanding academic performance and active participation in Youth Union activities.

RESEARCH EXPERIENCE

Center for Computational Science @HNUE

Research Assistant

Hanoi, Vietnam

Sep 2025 – present

Project: Tuning the Emission and Circularly Luminescence Dissymmetry of CzBN-Based Heterohelicenes

(under review – 3rd author)

Supervisor: Professor Hue Minh Thi Nguyen

- Performing density functional theory (DFT) calculations using Gaussian and VASP for research projects; preparing figures, tables, and visualizations for reports, academic papers, and presentations.
- Performing statistical analyses and interpret results to support research hypotheses.

Lumiere Research

Mar 2026 – Jun 2026

Project: Evaluating surface functionalization strategies in carbon nanotube-based drug delivery for cancer therapy: A review

Advisor: Dr. Simran Keraï

- 100% financial aid; conducted a systematic literature review on carbon nanomaterials in drug delivery, biosensing, and biomedical diagnostics; analyzed structure–property relationships and functionalization strategies to improve nanomaterial biocompatibility.

SKILLS

Technical Skills: Computational Chemistry (Gaussian 09, VASP (beginner), Chemcraft); Programming (JavaScript, Python, C++); Development Tools (Git, GitHub, VS Code); Graphic Design (Figma).

Other Skills: Academic tutoring, Piano performance (intermediate level, 10 years of experience), Photography, Writing (creative).

Language: Vietnamese (native); English (IELTS C1); Mandarin (TOCFL A1).

TEACHING ASSISTANT

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Teaching Assistant

Hanoi, Vietnam

Jul 2025 – present

- Developing supplemental study materials and practice problems to reinforce learning; providing targeted tutoring on AP Chemistry topics, including kinetics, equilibrium, and electrochemistry.
- Supporting teachers in attendance tracking, grading, and answering students' questions.

LEADERSHIP & ACTIVITIES

- Founded Louisa's Artistry, a page sharing photography knowledge and photos. Nov 2024 – present
- Founded ChemOlyms, a student-led chemistry mentorship focused on Olympiad training in Vietnam. Apr 2025 – Jul 2026
- Lead Pianist, School Band - responsible for arranging and performing in live shows. Aug 2023 – Jul 2026
- Designed and taught chemistry for underprivileged children (Chemistry Instructor, Phat Tich Center). Jun 2023 – Aug 2024

ADDITIONAL AWARDS/HONORS

- Gold Medal (Rank 3rd), Asian Chemistry Olympiad (2025)
- Bronze Medal, Canadian Chemistry Contest (CCC) (2025)
- Gold Medal, C3L6 by University of Cambridge (2025)
- High Distinction, International Chemistry Quiz (ICQ) (2025)
- Silver Medal, Chemico (Stemco by NUS) (2025)
- Bronze Medal, Hanoi University of Science Olympiad (HSGSO) (2024)
- LHP Talent Scholarship (2025)
- Merit Award, Vietnam National Science – Chemistry Olympiad (2024)
- Second Prize, Provincial Excellent Student Competition in Chemistry (Grade 12, taken as an 11th grader, 2024)
- First Prize, Le Hong Phong Chemistry Research Conference (2025)